

EduForge

Coaching Institute Management Platform with Question Paper Generation

Version 1.0 – MVP Proposal

1. Executive Summary

EduForge is a cloud-based Coaching Institute Management Platform designed for small and medium coaching institutes, tuition centers, and educational organizations.

The platform aims to digitize daily operations such as:

- Student Management
- Attendance Tracking
- Fee Management
- Question Bank Management
- Question Paper Generation

The primary objective is to reduce manual work, improve operational efficiency, and provide teachers with a simple way to generate examination papers.

2. Problem Statement

Most coaching institutes currently depend on:

- Excel Sheets
- WhatsApp Groups
- Physical Registers
- Manual Fee Records
- Word Documents for Question Papers

This creates several challenges:

Administrative Challenges

- Student records are scattered.
- Attendance tracking is manual.
- Fee collection monitoring is difficult.
- Batch management becomes complex as student count grows.

Academic Challenges

- Question papers are created manually.
 - Question repetition occurs frequently.
 - Difficulty balancing paper complexity.
 - Significant teacher effort is spent preparing exams.
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3. Proposed Solution

EduForge will provide a centralized web application enabling institutes to manage their complete academic and operational workflow.

The platform will provide:

Administration

- Student Registration
- Batch Management
- Attendance Management
- Fee Tracking

Examination Management

- Question Bank
 - Question Categorization
 - Automated Question Paper Generation
 - PDF Export
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4. Target Audience

Primary Customers

- Tuition Teachers
- Coaching Institutes
- Competitive Exam Centers
- Small Educational Organizations

Initial Market Focus

- CBSE Coaching Centers
 - State Board Coaching Institutes
 - Tuition Classes (Class 6–12)
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5. Product Features (Phase 1 MVP)

Module 1: Authentication & User Management

Features

- Login
- Logout
- Password Change
- Role-Based Access

Roles

- Admin
 - Teacher
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Module 2: Student Management

Features

- Add Student
- Edit Student
- Delete Student
- Search Student
- Batch Assignment

Student Information

- Student Name
 - Parent Name
 - Mobile Number
 - Class
 - Batch
 - Address
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Module 3: Attendance Management

Features

- Daily Attendance
- Attendance History
- Monthly Attendance Reports

Attendance Status

- Present
 - Absent
 - Late
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Module 4: Fee Management

Features

- Fee Collection
- Due Tracking
- Payment History
- Pending Fee Reports

Dashboard Metrics

- Total Fees Collected
 - Pending Fees
 - Student-wise Fee Status
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Module 5: Question Bank Management

Teachers can maintain a reusable repository of questions.

Features

- Add Question
- Edit Question
- Delete Question
- Search Question

Categorization

- Subject
- Chapter
- Marks
- Difficulty Level

Difficulty Levels

- Easy
 - Medium
 - Hard
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Module 6: Question Paper Generator

User Input

- Class
- Subject
- Chapters
- Total Marks
- Difficulty Distribution

System Output

- Auto Generated Question Paper
- Balanced Marks Distribution
- PDF Export

Benefits

- Saves Teacher Time
 - Reduces Repetition
 - Standardizes Exam Papers
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Module 7: Dashboard

Dashboard Information

- Total Students
 - Active Batches
 - Attendance Percentage
 - Fee Collection Summary
 - Upcoming Examinations
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6. Technical Architecture

Frontend

Technology Stack:

- Angular
- Angular Material
- Bootstrap

Responsibilities:

- User Interface
 - Form Validation
 - Dashboard Visualization
 - PDF Download Interface
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Backend

Technology Stack:

- Java 21
- Spring Boot 3
- Spring Security
- JWT Authentication
- Spring Data JPA

Responsibilities:

- Business Logic
 - Authentication & Authorization
 - Question Paper Generation
 - Report Processing
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Database

Technology:

- MySQL 8

Core Tables:

- Users
 - Students
 - Attendance
 - Fees
 - Subjects
 - Questions
 - Question Papers
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PDF Generation

Technology:

- OpenPDF

Capabilities:

- Generate Question Papers
 - Institute Branding
 - Downloadable PDF Format
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7. High-Level Architecture

User



Angular Web Application



Spring Boot REST APIs



Business Modules

- Authentication
- Student Management
- Attendance Management
- Fee Management
- Question Bank Management
- Question Paper Generation



MySQL Database



PDF Generation Engine



Question Paper Download

8. Non-Functional Requirements

Security

- JWT Authentication
- Password Encryption
- Role-Based Access

Performance

- Support up to 100 Coaching Institutes
- Support up to 10,000 Students

Availability

- 99% Application Availability Target

Scalability

- Modular Monolithic Architecture
 - Future Ready for Expansion
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9. Hosting & Infrastructure Cost (Phase 1)

Domain

Custom Domain

Example:

eduforge.in

Estimated Cost:

₹800 – ₹1,200 per year

Frontend Hosting

Platform:

- Vercel or
- Netlify

Estimated Cost:

₹0/month

Backend Hosting

Platform:

- Render

Estimated Cost:

₹0 – ₹500/month

Database Hosting

Platform:

- MySQL (Cloud Hosted)

Estimated Cost:

₹0/month (Initial Free Tier)

SSL Certificate

Platform:

- Let's Encrypt

Estimated Cost:

Free

10. Estimated MVP Cost

Component	Estimated Cost
Domain	₹1,200/year
Frontend Hosting	Free
Backend Hosting	₹0-₹500/month
Database	Free
SSL	Free

Estimated First-Year Cost

₹4,000 – ₹7,000

11. Development Roadmap

Week 1

- Project Setup
- Authentication Module

Week 2

- Student Management

Week 3

- Attendance Management

Week 4

- Fee Management

Week 5

- Question Bank Module

Week 6

- Question Paper Generator
- PDF Export

Week 7

- Dashboard
- Reporting

Week 8

- Deployment
 - User Testing
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12. Future Enhancements (Phase 2)

- Parent Portal
 - SMS Notifications
 - WhatsApp Integration
 - Online Fee Payments
 - Mobile Application
 - Exam Analytics
 - AI-Assisted Question Generation
 - Multi-Institute Support
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13. Success Metrics

First 3 Months

- 5 Coaching Institutes Onboarded
- 500+ Students Managed
- 100+ Question Papers Generated

First Year

- 50+ Coaching Institutes
 - Recurring Subscription Revenue
 - Product-Market Validation
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Conclusion

EduForge aims to provide a simple, affordable, and scalable platform for coaching institutes by combining operational management with automated question paper generation. The MVP focuses on solving everyday institute challenges while maintaining low development and operational costs.